College of Business
香港城市大學
City University of Hong Kong

FinTech and AI in Finance

Class 3: Linear Machine Learning for Return Prediction

Gavin Feng

Semester A, 2026/27

City University of Hong Kong

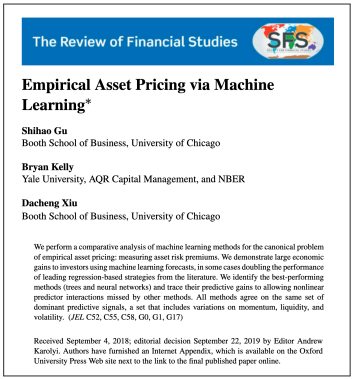
Market Return Prediction and Stock Selection

Two forecasting tasks, with different targets and benchmarks:

- **Monthly market excess returns:** macro predictors; lagged MA(12) benchmark. Compare OLS, forward selection, Ridge, LASSO, and Elastic Net.
- **Weekly stock excess returns:** technical characteristics; zero benchmark. Add PCR and PLS, then compare stock rankings.

Market Prediction with Linear ML

A Landmark Paper for Machine Learning in Finance



The Review of Financial Studies

Empirical Asset Pricing via Machine Learning*

Shihao Gu
Booth School of Business, University of Chicago

Bryan Kelly
Yale University, AQR Capital Management, and NBER

Dacheng Xiu
Booth School of Business, University of Chicago

We perform a comparative analysis of machine learning methods for the canonical problem of empirical asset pricing: measuring asset risk premiums. We demonstrate large economic gains to investors using machine learning forecasts, in some cases doubling the performance of leading regression-based strategies from the literature. We identify the best-performing methods (trees and neural networks) and trace their predictive gains to allowing nonlinear predictor interactions missed by other methods. All methods agree on the same set of dominant predictive signals, a set that includes variations on momentum, liquidity, and volatility. (*JEL* C52, C55, C58, G0, G1, G17)

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Gu, Kelly, and Xiu (2020)

- Estimate risk premia as a prediction problem.
- Linear shrinkage, trees, and neural networks face the same data and OOS test.
- Evaluate return-forecast accuracy and portfolio performance.

The course sequence

Class 3: selection, shrinkage, and dimension reduction.
Class 4: nonlinear interactions, trees, ensembles, and neural networks.

Class 5 examines characteristic importance and portfolio performance.

Four U.S. Macro Predictors across Three Equity Markets

Predictor	Forecast lag	Economic reading
Term spread	one month	monetary and business-cycle conditions
Federal funds rate	one month	current policy stance
Unemployment rate	two months	labor-market slack with release delay
Inflation rate	two months	price pressure with release delay

Same U.S. predictors for S&P 500, Hang Seng, and CSI 300 monthly excess returns. Class 2 used Chinese macro data for its CSI 300 example.

Ordinary Least Squares: A Monthly Market Excess-Return Forecast

OLS chooses the intercept and slopes to minimize training squared error:

$$\mathcal{L}(\alpha, \beta) = \frac{1}{2n} \sum_{t \in \mathcal{S}} \left(r_t^e - \alpha - z_t' \beta \right)^2, \qquad (\widehat{\alpha}, \widehat{\beta})_{\mathrm{OLS}} = \arg\min_{\alpha, \beta} \mathcal{L}(\alpha, \beta).$$

Here $\mathcal{S}$ contains n training months; z_t holds standardized inputs known before month t.

$$\widehat{r}_t^e = \widehat{\alpha} + \sum_{j=1}^{p} \widehat{\beta}_j z_{jt}.$$

OLS: no slope penalty. Correlated predictors can make coefficients unstable.

OLS: Calculating Next Month's Market Excess-Return Forecast

Fit the coefficients on training months. Suppose the market excess-return rule is

$$\widehat{r}^{e} = 0.20\% + 0.40\%\, z_{\text{spread}} - 0.30\%\, z_{\text{rate}},$$

where each z is measured in training-sample standard deviations.

For next month's forecast, $z_{\text{spread}} = 0.5$ and $z_{\text{rate}} = -1$:

$$\widehat{r}^{e} = 0.20\% + 0.40\%(0.5) - 0.30\%(-1) = 0.70\%.$$

Transform new inputs using the training means and standard deviations.

Illustrative coefficients and inputs; not estimates from the course dataset.

Freeze the Market Chronology before Comparing Models



OLS clarification: OLS has no tuning parameter. Its validation loss is used only for specification or model comparison.

After choosing the settings, refit on all 178 pre-test months. Two initial months are used to construct the lagged predictors.

Keep in mind

All three markets use the same predictors, dates, loss, and lagged MA(12) excess-return benchmark for OOS R^2.

Selection and Shrinkage

Forward Selection: Building a Short Predictor List

Begin with an intercept-only model.

1. Add each unused predictor in turn. Fit every candidate by OLS on the same training months.
2. Compare validation MSE. Keep the best candidate only if it improves on the current model.
3. Repeat with the remaining predictors. Stop when validation MSE no longer falls.

The result is a short predictor list. Signals useful only together may be missed.

Worked Example: Forward Selection Stops when Validation Error Rises

Step	Candidate model	Validation MSE $\times 10^4$	Decision
0	intercept only	30.00	starting point
1	add term spread	28.70	keep
2	add inflation	**28.20**	keep
3	add unemployment	28.40	stop

Keep in mind

The third predictor improves development-sample fit but worsens validation error. The two-predictor model is frozen.

Illustrative values for mechanics; not empirical results from the course dataset.

Shrinkage in Market Excess-Return Prediction

Estimate a linear forecast by minimizing training error plus a penalty on large slopes:

$$\underbrace{\mathcal{L}(\alpha, \beta)}_{\text{training squared error}} + \lambda \underbrace{P(\beta)}_{\text{slope penalty}} .$$

- $\lambda = 0$ gives OLS. Larger λ puts more weight on small slopes.
- Standardize predictors using training data and leave the intercept unpenalized.
- Choose λ on validation data. Too much shrinkage can remove useful signal.

The forecast remains linear: $\widehat{r}^e = \widehat{\alpha} + z'\widehat{\beta}$.

Ridge Regression: Keeping Related Signals

Ridge adds a squared-slope penalty to the OLS loss $\mathcal{L}$:

$$(\widehat{\alpha}, \widehat{\beta})_{\text{Ridge}} = \arg\min_{\alpha,\beta} \left\{ \mathcal{L}(\alpha, \beta) + \underbrace{\frac{\lambda}{2} \sum_{j=1}^{p} \beta_j^2}_{\text{Ridge penalty}} \right\}.$$

Larger slopes pay a quadratic cost. Larger λ means stronger shrinkage.

Use: overlapping predictors, such as the funds rate and term spread.

Effect: stabilize coefficients; generally retain all predictors.

Ridge: Choosing the Penalty and Making Predictions

1. Standardize the predictors using the 142 development months.
2. Fit Ridge at several penalty values; compare market excess-return forecasts on the next 36 validation months.
3. Refit the selected specification on all 178 pre-test months, then forecast the 60 test months.

In our U.S. example, validation selects software penalty `alpha`$= 100$:

$$\text{validation MSE} \times 10^4 = 27.02, \qquad R^2_{\text{OOS}} = -3.2\%.$$

Lower test loss than OLS; still worse than lagged MA(12).

With $\mathcal{L} = \text{SSE}/(2n)$ above, Ridge's software `alpha`$= n\lambda$; penalty scales differ across implementations.

LASSO: Selecting Signals through Shrinkage

LASSO replaces the squared-slope penalty with absolute values:

$$(\widehat{\alpha}, \widehat{\beta})_{\text{LASSO}} = \arg\min_{\alpha,\beta} \left\{ \mathcal{L}(\alpha, \beta) + \underbrace{\lambda \sum_{j=1}^{p} |\beta_j|}_{\text{LASSO penalty}} \right\}.$$

The absolute-value penalty has a corner at zero: some fitted slopes become exactly zero.

Effect: $\widehat{\beta}_j = 0$ removes predictor j from the forecast.

Caution: selection among correlated predictors can change across samples.

LASSO: Reading the Selected Market Excess-Return Forecast

Example coefficients after standardization and validation:

	Term spread	Funds rate	Inflation
OLS coefficient	0.60	0.50	0.20
LASSO coefficient	0.48	0	0

With intercept 0.10% and standardized spread 0.5:

$$\widehat{r}^{e} = 0.10\% + 0.48\%(0.5) = 0.34\%.$$

The other inputs contribute zero. With a smaller penalty, more signals may remain.

Illustrative coefficients in percentage points per standard deviation; not course estimates.

Elastic Net: Selection with Correlated Predictors

Elastic Net adds both penalties to the same OLS loss:

$$(\widehat{\alpha}, \widehat{\beta})_{\mathrm{EN}} = \arg\min_{\alpha,\beta} \left\{ \mathcal{L}(\alpha, \beta) + \lambda \left[\underbrace{\rho \sum_{j=1}^{p} |\beta_j|}_{\text{LASSO part}} + \underbrace{\frac{1-\rho}{2} \sum_{j=1}^{p} \beta_j^2}_{\text{Ridge part}} \right] \right\}.$$

$\lambda \geq 0$ controls total shrinkage; $0 \leq \rho \leq 1$ is the LASSO share.

$$\rho = 1: \text{ LASSO}, \qquad \rho = 0: \text{ Ridge}, \qquad 0 < \rho < 1: \text{ a mixture}.$$

Use it when a compact rule should retain several related signals. Choose both λ and ρ on validation data.

Elastic Net: Choose Both Settings on Validation Data

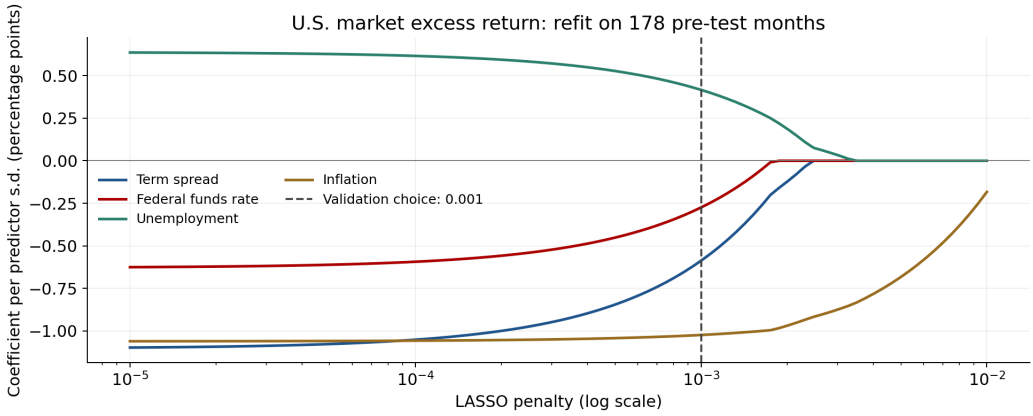
Fit each candidate on development months. Example validation MSE $\times 10^4$:

LASSO share	Penalty 0.0001	Penalty 0.001	Penalty 0.01
0.25	33.4	**26.8**	27.6
0.75	34.1	27.2	28.9

Select penalty 0.001 and LASSO share 0.25, then refit and forecast.

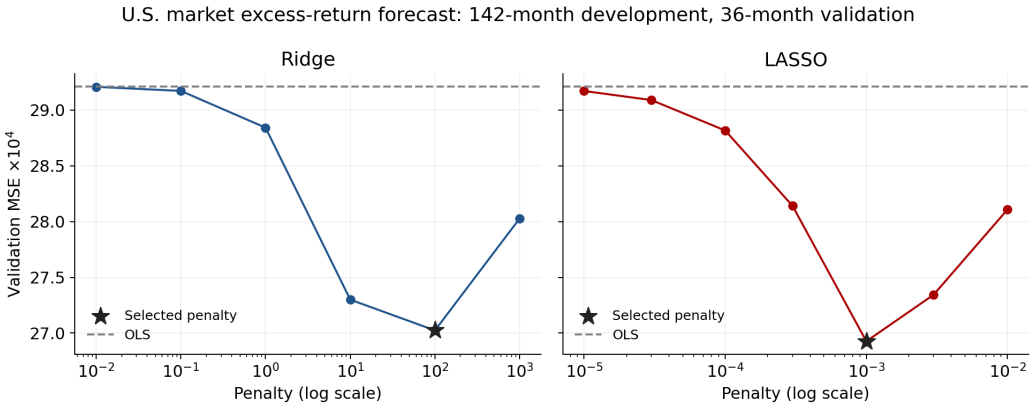
Illustrative validation grid, not empirical results from the course dataset.

How LASSO Coefficients Change with the Penalty



Validation selects a penalty that keeps all four slopes nonzero: shrinkage without variable removal.

Choose the Penalty on Validation Data



Fit each candidate on 142 development months and compare it on 36 validation months.

Worked Example: Validation Chooses U.S. LASSO

Using the four macro predictors and the 36-month validation period:

Model	Selected penalty	Validation MSE $\times 10^4$
OLS (kitchen sink)	–	29.21
Ridge	100.00	27.02
LASSO	0.001	**26.93**
Elastic Net	0.001	27.15

The loss differences are small; the table shows the model selected before the 60-month test. Course calculation using the 2006–2026 market-prediction sample.

Calculation Question: Evaluate the Market Excess-Return Forecast

For a **market excess-return** forecast, the test-period sum of squared errors is 0.018. The lagged MA(12) benchmark has a sum of squared errors of 0.015. Use

$$R^2_{\text{OOS}} = 1 - \frac{\text{SSE}_{\text{model}}}{\text{SSE}_{\text{benchmark}}}.$$

1. Calculate R^2_{OOS} and show the substitution.
2. State whether the model improves on the MA(12) benchmark.
3. Explain why MA(12) must be lagged rather than computed with future returns.

Be ready to explain

What does the sign of OOS R^2 tell a portfolio manager about forecast value?

The Same Validation Rule Meets the Test in Three Markets

Test OOS R^2 against lagged MA(12) excess returns over the same 60 months, after the refit:

Model	U.S.	HKSAR	Chinese Mainland
OLS (kitchen sink)	-22.1%	-13.7%	-148.2%
Ridge	-3.2%	+6.1%	-23.9%
LASSO	-16.3%	+10.1%	-0.6%
Elastic Net	-17.6%	+8.4%	-18.3%

Keep in mind

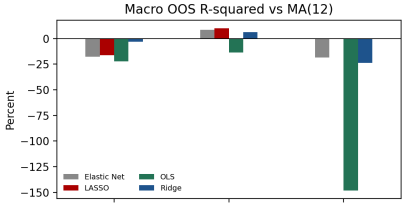
Validation choices: U.S.—LASSO; HKSAR—Elastic Net; Chinese Mainland—LASSO.

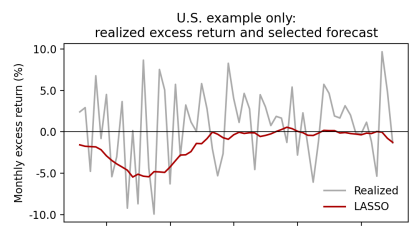
The selected U.S. and Chinese mainland models lose to MA(12); HKSAR beats it. Do not switch to HKSAR LASSO after seeing its higher test score.

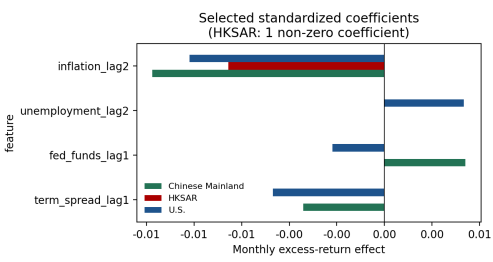
Market Excess-Return Forecasting Is Difficult in All Three Markets

Class 3 starts with a four-predictor macro-to-market problem









Let Codex Fit Ridge and LASSO under One Specification

```
candidates = {"ridge": Ridge(alpha=100),
              "lasso": Lasso(alpha=0.001, max_iter=100000)}

for name, estimator in candidates.items():
    model = make_pipeline(StandardScaler(), estimator)
    model.fit(train[x_names], train["market_excess_return"])
    forecast[name] = model.predict(test[x_names])
```

The pipeline fits scaling on `train` and reuses it on `test`. The displayed penalties were chosen on the 36 validation months. In the software, `alpha` denotes penalty strength.

Multiple-Choice Question: Choosing a Market Excess-Return Rule

Candidate models use the same macro predictors and chronological split. Their validation MSEs are:

Model	OLS	Ridge	LASSO	Elastic Net
Validation MSE	0.0025	0.0020	0.0018	0.0019

Under selection by validation MSE, what should happen next?

A) Freeze the LASSO settings, refit on development plus validation data, and evaluate on the test period.

B) Choose OLS because an unrestricted model uses the most information.

C) Compare all test errors first, then decide which model to report as selected.

D) Increase the LASSO penalty until the test-period forecast error falls.

Multiple-Choice Question: Predictor Units and Penalties

An analyst changes inflation from decimal units (0.03) to percentage units (3), but otherwise keeps the observations unchanged. A LASSO model uses the same penalty.

Which procedure makes the penalty comparison meaningful across predictor units?

A) Use means and standard deviations calculated from all dates, including the test period.

B) Leave predictors unscaled because the same penalty automatically removes unit effects.

C) Re-estimate the scaling separately on the validation sample before comparing forecasts.

D) Standardize using development-sample means and standard deviations, then apply those values to validation data.

Stock Returns and Characteristics

The Observation Changes from Month to Stock-Week

Which stocks will earn higher excess returns next week?

Element	Market problem	Stock problem
One row	one month	one stock in one forecast week
Target	market excess return	stock excess return over the risk-free rate
OOS benchmark	lagged MA(12) excess return	zero excess return
Predictors	lagged macro variables	30 lagged technical ranks
Sample	238 months	12,480 grid rows; 12,281 usable rows
Validation	chronological months	chronological weeks, all stocks together

Keep in mind

Same linear methods; new target, frequency, and benchmark.

A Fixed July 2023 Stock Universe

- The 12,480-row grid covers 2023-07-07–2026-06-26: 156 weeks $\times$ 80 Hang Seng Index constituents.
- Membership is fixed at the start of the exercise, rather than selected using the June 2026 list.
- Missing returns remain missing. Later index additions do not replace the original members.

This follows a fixed stock sample through time, not the changing composition of an index-tracking fund.

Six Indicator Families Organize the Thirty Columns

Family	Beginner examples	Question the signal asks
Trend	20-day momentum; close/SMA	has the price been rising?
Bar geometry	close-location value	where did the close fall inside the day's range?
Activity	log dollar volume	how much money traded?
Volatility	realized volatility	how variable were recent returns?
Oscillator	RSI	were recent gains unusually persistent?
Market risk	60-day beta	how strongly did the stock move with its market?

For example, 20-day momentum is $C_t / C_{t-20} - 1$. Weekly cross-sectional ranks put indicators measured in different units on a common scale.

Daily Data Become a Weekly Forecast Row without Leakage

For stock i in forecast week t:

$$\{O, H, L, C, V\}_{i,\,\text{through week } t-1} \longrightarrow x_{i,t-1} \longrightarrow \text{rank}_{m,t-1}(x_i) \longrightarrow \widehat{r}^{e}_{i,t} \longrightarrow r^{e}_{i,t}.$$

Ranks are computed inside the Hang Seng Index cross-section. We use

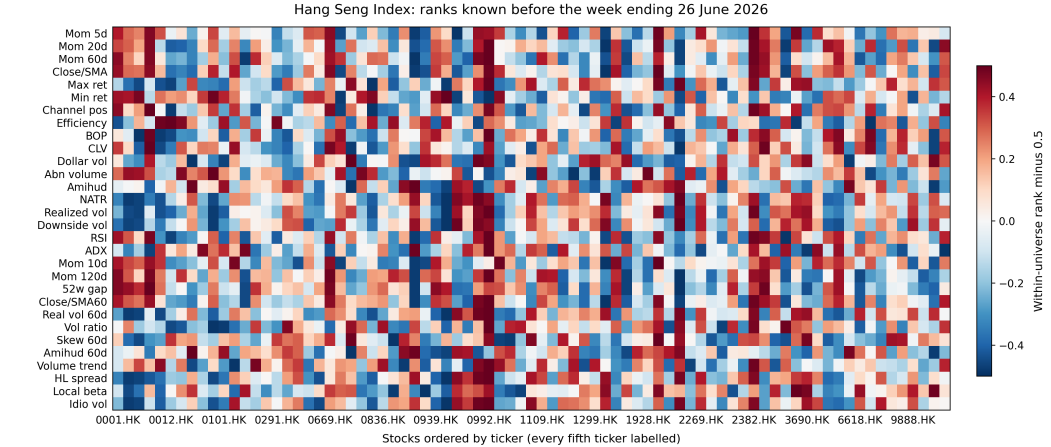
$$\text{centered rank} = \text{within-universe percentile rank} - 0.50,$$

so zero represents the middle of the cross-section. The raw indicators use only prices and volumes available before the forecast week.

No leakage

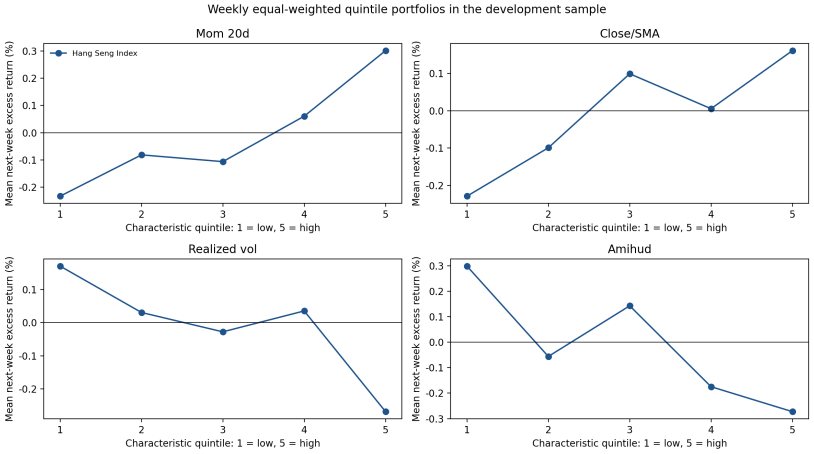
The week ending 26 June 2026 uses characteristics through the prior week. Its 19–26 June return is an outcome, not an input.

Thirty Ranks Describe One Weekly Cross-Section



Each column is a stock; each row is a characteristic. Colors show ranks within the index, not raw units.

Characteristic-Sorted Quintile Portfolio Returns



Each development week forms five HSI stock portfolios from one lagged characteristic. Points report the time-series mean of the equal-weighted next-week excess return.

Estimate One Linear Rule for the Cross-Section

Fit one coefficient vector across stock-week observations; z contains training-standardized characteristics:

$$\widehat{r}^{e}_{i,t} = \widehat{\alpha} + z'_{i,t-1}\widehat{\beta}.$$

Block	Forecast weeks	Weeks	Usable rows
Development	2023-07-07–2024-12-27	78	6,123
Validation	2025-01-03–2025-06-27	26	2,050
Refit	2023-07-07–2025-06-27	104	8,173
Test	2025-07-04–2026-06-26	52	4,108

Keep in mind

Seventy-nine of the eighty members supply outcomes. Hang Seng Bank contributes none, and those rows are left missing rather than imputed.

Multiple-Choice Question: Which Linear Tool Fits the Job?

A desk has 20 predictors in correlated valuation, trend, and liquidity families. It wants a **compact rule that retains related signals together**. All three models are within one standard error of the lowest validation MSE.

Model	Validation MSE	Nonzero	Pattern
Ridge	0.0102	20	all signals retained
LASSO	0.0100	4	some families represented once
Elastic Net	0.0101	9	several signals per family

Which choice best matches the stated financial job?

A. Ridge, because retaining every signal is always safest.

B. LASSO, because the smallest validation MSE must dominate.

C. Elastic Net, because it balances sparsity and grouped retention.

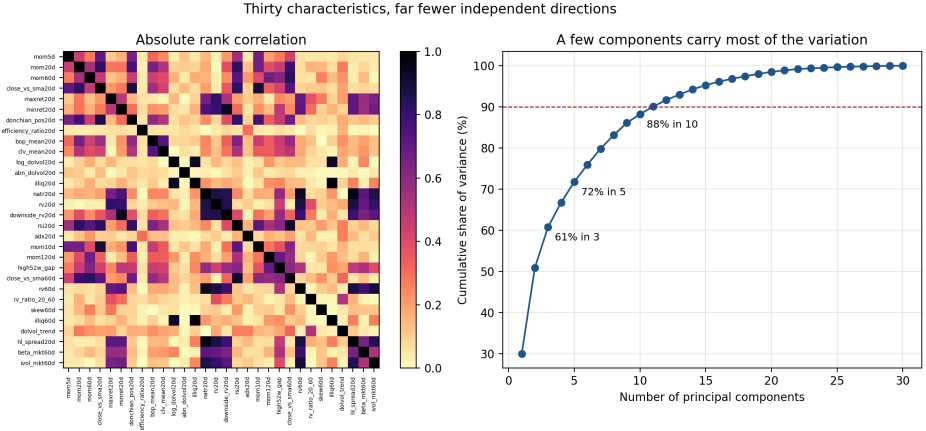
D. Choose whichever model has the lowest frozen-test MSE.

Discuss

Why should a 0.0001 validation-MSE difference not override the implementation goal?

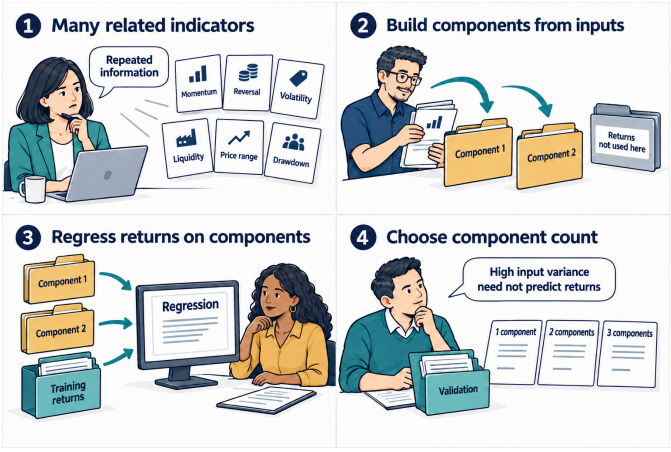
Dimension Reduction

Thirty Characteristics Are Not Thirty Signals



Mean absolute rank correlation 0.25; 61% of the variance sits in three components and 88% in ten.

Principal Component Regression: Summarizing Many Inputs



When useful: related inputs carry overlapping information.

Form K components from training inputs, without using returns.

Fit OLS on these component scores:

$$\widehat{r}_i^e = \widehat{\alpha} + \sum_{k=1}^{K} \widehat{\gamma}_k c_{ik}.$$

Choose $K < p$ by validation error. High input variance need not predict returns.

PCR: Calculating a Stock Excess-Return Forecast

Suppose PCR summarizes the standardized indicators in two components:

$$PC_1 = 0.6 z_{\text{momentum}} + 0.8 z_{\text{MA gap}}, \qquad PC_2 = z_{\text{volatility}}.$$

The fitted weekly excess-return forecast is

$$\widehat{r}_i^e = 0.10\% + 0.30\%\, PC_{1,i} - 0.20\%\, PC_{2,i}.$$

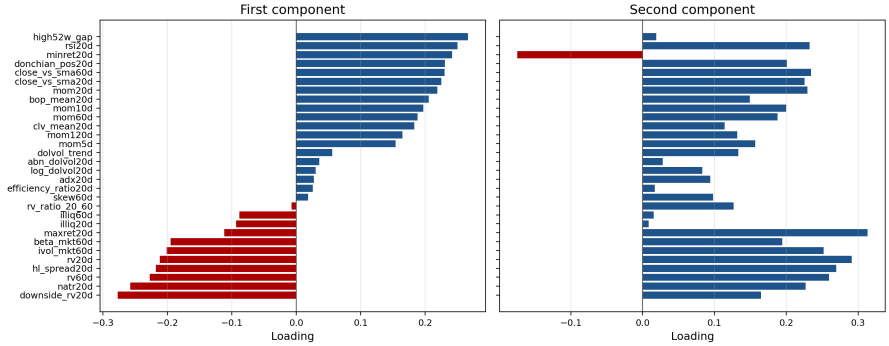
For $z_{\text{momentum}} = 1$, $z_{\text{MA gap}} = 0.5$, and $z_{\text{volatility}} = -0.5$:

$$PC_1 = 1.0, \quad PC_2 = -0.5, \quad \widehat{r}_i^e = 0.50\%.$$

For new stocks, apply the fitted scaling, component loadings, and return coefficients in that order.

Illustrative loadings and coefficients, not estimates from the course panel.

PCR: What a Component Is Made Of



Keep in mind

PC1 combines positive trend loadings with negative volatility loadings: calm uptrends versus volatile stocks.

Partial Least Squares: Components That Use Returns

PLS uses training excess returns r^e when choosing a component. For the first component:

$$w_1 = \arg \max_{\|w\|_2=1} \mathrm{Cov}_{\mathcal{S}}(Zw, r^e)^2, \qquad c_{i1} = z_i' w_1.$$

Z: standardized training predictors; r^e: training excess returns.

$$\widehat{r}_i^e = \widehat{\alpha} + \sum_{k=1}^{K} \widehat{\gamma}_k c_{ik}.$$

Extract later components from residual predictor and return information. Select K by validation error.

PLS uses both Z and r^e: returns guide the components; the forecast remains linear.

PLS: From a Component to a Stock Excess-Return Forecast

After fitting a one-component PLS model on standardized training data, suppose its component and forecast are

$$c = 0.8 z_{\text{momentum}} - 0.6 z_{\text{volatility}}, \qquad \widehat{r}^{e} = 0.10\% + 0.30\%\, c.$$

For a new stock with $z_{\text{momentum}} = 0.5$ and $z_{\text{volatility}} = -1$:

$$c = 0.8(0.5) - 0.6(-1) = 1.0, \qquad \widehat{r}^{e} = 0.40\%.$$

Choose the component count on validation weeks; refit, then freeze the transformation for testing.

Illustrative component weights and coefficients, not estimates from the course panel.

Weekly Prediction Evidence

Six Linear Models on the Same Stock Panel

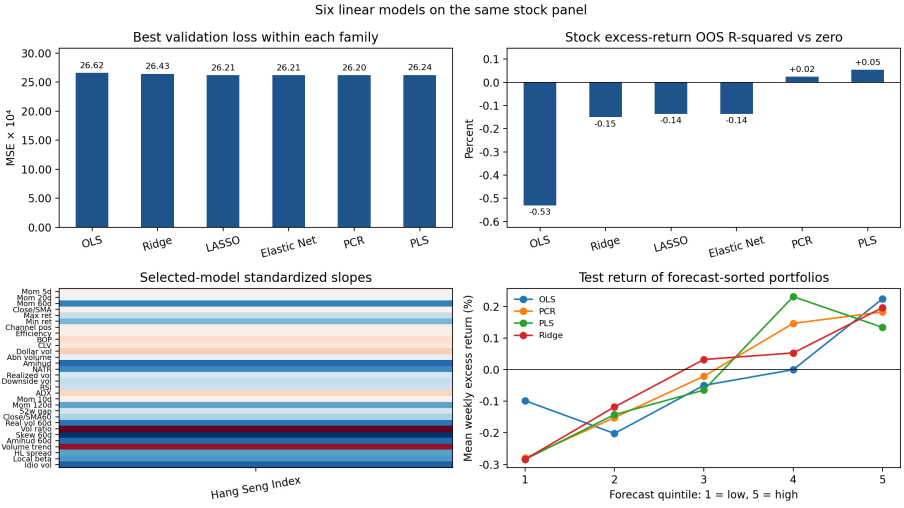
Model	Selected setting	Validation MSE $\times 10^4$	Test OOS R^2
PCR (selected)	5 components	26.20	+0.02%
LASSO	penalty 0.003	26.21	-0.14%
Elastic Net	penalty 0.01	26.21	-0.14%
PLS	1 component	26.24	+0.05%
Ridge	penalty 1000	26.43	-0.15%
OLS	unrestricted	26.62	-0.53%

Validation selects PCR with five components. PCR and PLS have test OOS R^2 only slightly above zero.

These small point estimates do not establish persistent improvement.

Hang Seng Index panel. Test: 52 weeks, 4,108 stock-week observations. Benchmark: zero excess return.

Linear Excess-Return Forecasts Remain Weak



Three Stock Forecasts from One Test Week

For the week ending 4 July 2025, the validation-selected PCR model produces:

Stock	Forecast $\widehat{r}^e_{i,t}$	Realized $r^e_{i,t}$	Reading
CCB (0939.HK)	+0.85%	+2.16%	direction and magnitude close
Meituan (3690.HK)	-0.07%	-6.73%	correct direction, small forecast
Xinyi Solar (0968.HK)	-0.24%	+20.42%	wrong direction, extreme move

Keep in mind

The forecasts stay near zero while realized weekly returns can be much wider. Model evaluation uses all 4,108 stock-week forecasts.

All figures are excess returns. The period-matched risk-free return was +0.08%, subtracted before rounding.

Vibe Coding: Compare Excess-Return Forecasts

Instruction

Using the weekly stock panel and the stated date split, compare OLS, Ridge, LASSO, Elastic Net, PCR, and PLS. Select settings on validation weeks, refit through June 2025, and report test OOS R^2 against zero excess return.

Ask for a second output: the predicted and realized excess return of one named stock in one test week.

Read the model comparison first, then trace the named stock through the fitted rule.

A Constant Excess-Return Forecast Cannot Rank Stocks

In this stock panel, the selected penalties in LASSO and Elastic Net set all thirty slopes to zero:

$$\widehat{r}^{e}_{i,t} = \widehat{\alpha} \qquad \text{for every stock } i.$$

- Predicting near the training mean can be competitive under MSE.
- Identical excess-return forecasts provide no stock ranking.
- Breaking ties by ticker or row order creates an arbitrary signal.

Equal weighting remains feasible; this excess-return forecast supplies no stock ranking.

Prediction Accuracy and Stock Rankings

Sorting Stocks by Their Return Forecasts

1. Rank stocks by their predicted excess return before the holding week.
2. Assign five groups using those forecast ranks.
3. Calculate each group's realized excess return during the holding week.
4. Compare the top and bottom groups over all test weeks.

The sort asks whether higher forecasts identify better subsequent returns. Class 5 adds position sizes, portfolio constraints, and risk adjustment.

Return Forecast Magnitude and Stock Ranking

Two models can give the same ranking while making different numerical forecasts.

Stock	Realized excess return	Model A	Model B	Rank
A	0.1%	0.2%	0.4%	low
B	0.2%	0.4%	0.8%	middle
C	0.3%	0.6%	1.2%	high

Model B doubles every forecast. Its squared error is larger, but the high-minus-low return is the same: $0.3\% - 0.1\% = 0.2\%$.

MSE assesses forecast levels. A sort assesses ordering.

Illustrative values for one holding week.

Ridge: Weak Return-Forecast Accuracy, Positive Realized Spread

Ridge forecast	Q1	Q2	Q3	Q4	Q5	Q5−Q1
Mean weekly excess return	-0.28%	-0.12%	+0.03%	+0.05%	+0.20%	+0.48%

Ridge has OOS $R^2 = -0.15\%$ against zero excess return, but its realized top–bottom spread is positive.

Read the spread carefully

52 test weeks; $t = 1.46$.

Mean weekly spread: +0.13% in the first half; +0.83% in the second. Trading costs excluded.

Return-forecast accuracy and portfolio ranking measure different properties of the same forecasts.

The Question for Nonlinear ML

- Linear models use the same slope for a predictor across observations.
- Should momentum have the same effect for low- and high-volatility stocks?
- Class 4 tests whether thresholds and interactions improve these forecasts.